

Generation Interconnection System Impact Study Report

For Queue Project Q20-1617

Glen Oaks 138 kV

June 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1617. The Interconnection Customer proposes a 163.7 MW solar photovoltaic facility interconnecting at the Glen Oaks 138 kV substation in the Virginia area. The request was received on December 07, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 477 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,660,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: S Texa to Glen Oaks 138 kV
Contingency: Brush Tavern to Rustburg 138 kV
Loading: 139.4% (Rating: 246 MVA)
DFAX: 80.6%
Pre-project loading: 85.8%
2. Facility: Barksdale to Tangy 138 kV
Contingency: Clover to Bear 345 kV
Loading: 126.9% (Rating: 214 MVA)
DFAX: 6.2%
Pre-project loading: 122.1%
3. Facility: STexa to Franklin-Elk Grove 138 kV
Contingency: Desert Outpost to S Texa 138 kV
Loading: 119.6% (Rating: 292 MVA)
DFAX: 38.8%
Pre-project loading: 97.9%
4. Facility: New London to Thornhill 138 kV
Contingency: La Joie to Carson Ice Cogen 138 kV
Loading: 108.7% (Rating: 242 MVA)
DFAX: 7.0%
Pre-project loading: 104.0%
5. Facility: Thornhill to Carson Ice Cogen 138 kV
Contingency: Carson Ice Cogen to La Joie 138 kV Line
Loading: 106.0% (Rating: 234 MVA)
DFAX: 7.0%
Pre-project loading: 101.1%
6. Facility: Desert Outpost to S Texa 138 kV line
Contingency: Brush Tavern to Rustburg 138 kV
Loading: 105.5% (Rating: 503 MVA)

DFAX: 34.2%

Pre-project loading: 94.3%

7. Facility: Desert Outpost to Monel 138 kV

Contingency: Brush Tavern to Rustburg 138 kV

Loading: 105.4% (Rating: 379 MVA)

DFAX: 19.4%

Pre-project loading: 97.1%

8. Facility: Scottsville to Rock Island Dam 138 kV

Contingency: Reach to Bluff City 138 kV

Loading: 102.1% (Rating: 235 MVA)

DFAX: 7.1%

Pre-project loading: 97.1%

9. Facility: Monel to Candelaria Substation 138 kV

Contingency: Bear to Cloer 345 kV

Loading: 100.4% (Rating: 548 MVA)

DFAX: 11.4%

Pre-project loading: 97.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Glen Oaks to South Texa 138 kV	Rebuild line with higher-capacity conductor	\$2,295,000	50% (\$1,153,000)
Barksdale to Tangy 138 kV	Rebuild line with higher-capacity conductor	\$25,347,000	100% (\$25,347,000)
STexa to Franklin-Elk Grove 138 kV	Rebuild line with higher-capacity conductor	\$20,417,000	27% (\$5,476,000)
New London to Thornhill 138 kV	Reconductor line	\$6,992,000	100% (\$6,992,000)
Carson Ice Cogen to Thornhill 138 kV Line	Reconductor line	\$2,529,000	100% (\$2,529,000)
Desert Outpost to S Texa 138 kV line	Reconductor line	\$438,000	20% (\$87,000)
Desert Outpost to Monel 138 kV	Reconductor line	\$1,813,000	25% (\$454,000)
Rock Island Dam to Scottsville 138 kV	Reconductor line	\$5,610,000	60% (\$3,372,000)
Monel Substation to Candelaria 138 kV	Reconductor line	\$18,094,000	12% (\$2,250,000)
	Total allocated to project		\$47,660,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.